AI & DS Knowledge Assistant — Daily Engineering Timeline

MVP · 4 weeks (20 working days) · 3 team members · Day-by-day plan from scratch to final demo. Estimates, not commitments.

Week 0 · Pre-launch prep (all, ~0.5 day each): confirm the synopsis scope · collect syllabus + regulation PDFs · get Gemini API key · create GitHub repo and Streamlit Cloud account · agree on folder structure and data schema. **Exit:** nothing blocks Day 1.

Preethi
Frontend · Streamlit UI

- Navigation: Year → Semester → Subject → Module
- Chat interface with source references
- Summary, Questions and Quiz tabs
- Knowledge-network page, UI polish, usability testing

Pragna
Backend · RAG + Gemini

- PDF extraction, chunking, SBERT embeddings, vector index
- RAG chain with page-referenced answers
- Gemini summaries, questions and quizzes
- Caching, error handling, deployment

Nishant
Data + Network Analysis

- Collect and organise syllabus and regulation PDFs
- Evaluation set and retrieval accuracy checks
- NetworkX graph, centrality, Louvain communities
- Project report and references

Milestones: D5 data + navigation ready · D10 RAG chatbot closed · D15 all features working · D20 final demo + report

Week 1 · Foundations (data + skeleton)

Day	Preethi · Frontend	Pragna · Backend + AI	Nishant · Data + Network	Milestone / Exit
D1	Set up repo + Streamlit skeleton; sketch wireframes of all pages	Set up environment, Gemini key, requirements; choose embedding model + vector store	Collect syllabus and regulation PDFs; define data schema (Year → Sem → Subject → Module)	
D2	Sidebar navigation using stub data	PDF text extraction + chunking prototype	Clean syllabus into structured JSON/CSV	
D3	Module page layout with tabs: Summary, Q&A, Questions, Quiz	SBERT embeddings + vector index on a sample PDF	Parse regulation PDF into sections with page numbers	
D4	Connect navigation to real JSON data	Retrieval function: top-k chunks with page references	Finish data for all semesters; validate completeness	
D5	Chat UI shell (input, message list)	Gemini API wrapper + prompt templates	Draft concept list per subject (graph nodes/edges)	**D5 — Data organised; navigation works on real data; retrieval returns correct chunks**

Week 2 · Core Loop (RAG chatbots)

Day	Preethi · Frontend	Pragna · Backend + AI	Nishant · Data + Network	Milestone / Exit
D6	Wire chat UI to backend	RAG chain: retrieve → Gemini → answer	Index all regulation documents	
D7	Show source references and page numbers in answers	Citations in answers + "not found" fallback	Build per-module syllabus index	
D8	Chat history and session state	Syllabus Q&A mode (module-scoped retrieval)	Write 30 test questions with expected answers	
D9	Loading states, error messages, regulation page polish	Prompt tuning on failed cases	Measure retrieval accuracy; log failures	
D10	Integration: both chatbots in the UI; demo run	Fix retrieval/prompt issues from the run	Re-evaluate on 30-question set	**D10 — Core loop closed: both chatbots answer with page references; ≥80% correct on test set**

Week 3 · Module Intelligence + Network Analysis

Day	Preethi · Frontend	Pragna · Backend + AI	Nishant · Data + Network	Milestone / Exit
D11	Summary tab UI	Module summary generation (Gemini)	Build concept graph in NetworkX from syllabus	
D12	Questions tab (short + long answer)	Question generation endpoint	Degree and betweenness centrality	
D13	Quiz tab: MCQs, scoring, answer reveal	Quiz generation with JSON validation	Louvain community detection	
D14	Knowledge-network page (interactive graph)	Caching of outputs; API error and rate-limit handling	Export graph data + table of central/bridge concepts	
D15	Integrate network page; highlight top concepts	Connect all AI features end to end	Validate graph results; check quality of generated content	**D15 — All five features working end to end in one Streamlit app**

Week 4 · Harden & Launch

Day	Preethi · Frontend	Pragna · Backend + AI	Nishant · Data + Network	Milestone / Exit
D16	UI polish, consistent layout, friendly error messages	Optimise latency; fallback when API fails	Fill data gaps; extend test coverage	
D17	Usability test with 3–5 students	Fix backend bugs from testing	Review quiz/summary quality; collect feedback	
D18	Fix UI issues; capture screenshots for report	Deploy on Streamlit Cloud; secure API keys	Write report: methodology, system design, SNA results	
D19	Prepare demo script and presentation slides	Final regression testing on deployed app	Finish report: literature, results, references	
D20	Demo rehearsal (all three)	Demo rehearsal (all three)	Final submission package	**D20 — Final demo ready: deployed app + report + presentation**